ALFE COMMERCIAL SALES AND INVENTORY MANAGEMENT SYSTEM

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ABSTRACT

This project proposes a desktop-based sales and inventory management system for **ALFE Commercial**. The system is intended to organize product information, monitor inventory, record sales, and maintain transaction records in a centralized application. The proposed system will provide a more organized way of managing sales and inventory-related information. The desktop application is intended to use **Java Swing** for the user interface and core system functions, **MySQL** for data storage, and **phpMyAdmin** for database administration during development and maintenance.

The project will use an **Agile SDLC approach supported by Scrum practices**. Development will be divided into manageable stages so that the team can develop and test working features, collect feedback, and improve the system throughout development. The project documentation will also consider requirements, feasibility, possible risks, project schedule, estimated costs, and Scrum activities.

I. PROJECT BACKGROUND

1.1 Client / Organization Background

ALFE Commercial is a business that manages products, sales, and inventory records. The proposed system is intended to support the management of product information, available stock, sales, and transaction records through a centralized desktop application. The main user of the current system is the administrator, who can access the different management sections of the application.

The proposed **ALFE Commercial Sales and Inventory Management System** will provide a structured interface for accessing the Dashboard, Products, Inventory, Sales, and Transactions modules. The system is designed to organize important business records and make them easier to access and manage.

1.2 Current Process

The current project process focuses on managing product, inventory, sales, and transaction information. These activities require users to maintain accurate product details, monitor available stock, record sales information, and review transaction records.

The proposed system will organize these activities within a single desktop application. Users will be able to access the different modules through the system, enter and manage relevant information, and store data using a MySQL database.

1.3 Identified Problems

Based on the identified sales and inventory management needs, the following problems may occur when information is not managed through a centralized system:

· Product information may be difficult to organize and update when records are maintained separately.
· Monitoring available product stock may require additional effort.
· Sales and transaction information may be more difficult to review when records are not centralized.
· Retrieving specific product or transaction information may take additional time.

· Maintaining accurate and consistent records can become more difficult as the amount of information increases.

II. PROJECT PURPOSE AND OBJECTIVES

2.1 Project Purpose

The purpose of the project is to create a simple desktop-based sales and inventory management system that can help ALFE Commercial organize product information, monitor inventory, record sales, and manage transaction records.

The system aims to provide a centralized application where relevant information can be stored and accessed in an organized manner while providing users with a clear and consistent interface for performing sales and inventory-related tasks.

2.2 Main Project Objective

To design and develop a desktop-based **Sales and Inventory Management System for ALFE Commercial** that supports organized management of products, inventory, sales, and transaction records.

2.3 Specific Objectives

· To allow the administrator to securely access the system using login credentials.
· To manage product information such as product ID, name, brand, category, price, quantity, and status.
· To monitor and update product inventory and stock information.
· To record sales information, including selected products, quantities, and total amounts.
· To maintain and review transaction records through the Transactions module.
· To provide a centralized MySQL database for storing and retrieving system information.
· To provide a clear and user-friendly interface with consistent navigation between system modules.
· To apply appropriate IT project management practices throughout the planning, development, testing, and documentation of the project.

3. PROJECT SCOPE AND LIMITATIONS

3.1 Project Scope

User / Area	Included Functions
Administrator	Login, navigate the Home dashboard, access the Products, Inventory, Sales, Transactions, and Logout.
Products	Product name, brand , category, price, and quantity fields. Add Product, Update, Clear, Search, and Back interface controls.
Dashboard	Provides a central navigation area and summary information for system operations.
POS Terminal and Checkout	Supports selectiong products for a sale, entering quantities, calculating subtotals and totals, recording payment details, and calculating change.
Inventory and Stock History	Displays product stock information and support stock-related operations and review of stock history, including stock-in, stock-out, adjustment, and sale-related changes.
Sales and Reports	Provides sales reporting and data-based summaries. Cashier performance reporting is included in the project source.
Transaction	Transaction History, transaction ID/date/product/quantity/price/total/status information, transaction details, new transaction field, Add Transaction, Clear, Search, and Back controls.
Database	JDBC connection to the alfe_commercial database and authentication data in the users table.

3.2 Project Limitations

- The SQL database currently contains only the users table and administrator record.
- This system does not include online ordering, web access, or third party payment gateway integration.
- The system is intended for local business operations and is not designed for remote access.
- The system does not provide automatic notifications for low stock products.
- The system does not include barcode scanning for product identification.

4. PROJECT FEASIBILITY AND PROJECT CHARTER SUMMARY

4.1 Technical Feasibility

The ALFE Commercial Sales and Inventory Management System is technically feasible because the project uses commonly available software development tools and technologies. The system will be developed as a Java Swing desktop application using NetBeans IDE. MySQL will be used as the database for storing system information, while XAMPP and phpMyAdmin will be used for local database setup and administration during development.

The system's main functions include user login, product management, inventory monitoring, sales recording, and transaction management. These functions can be implemented using the selected technologies without requiring specialized hardware.

The system can run on a standard computer capable of running Java and the required MySQL database environment.Therefore, the technical requirements of the project are achievable using the resources available to the development team.

4.2 Operational Feasibility

The project is operationally feasible because it is designed to support common sales and inventory management activities of ALFE Commercial. Instead of accessing different sources of information separately, the proposed system provides a centralized desktop application for managing products, inventory, sales, and transaction records.

The system includes a Dashboard, Products, Inventory, Sales, and Transactions modules that can be accessed through the application's navigation sidebar. A login screen is also included to control access to the system.

The interface is designed to be simple and organized so that users can understand the available functions and navigate between modules. Basic user guidance may be provided to help users become familiar with the system.

4.3 Economic Feasibility

The project is economically feasible because the main development tools used by the team are readily available and do not require major software expenses for the development of the system. Java, NetBeans, Java Swing, MySQL, XAMPP, and phpMyAdmin can be used without purchasing additional software licenses.

The main expected expenses are related to internet or data, transportation, printing and documentation, backup storage, and other miscellaneous project activities. The estimated project cost is **PHP 3,000**, although the actual expenses may vary depending on the resources already available to the team.

The expected benefits include better organization of product, inventory, sales, and transaction information, as well as easier access to records through a centralized desktop application. Based on the available resources and estimated expenses, the project can be developed within a reasonable budget for a student project.

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4.4 Project Charter Summary

Charter Item	Project Entry
Project Title	ALFE Commercial Sales and Inventory Management System
Project Purpose	Improve the organization of product, inventory, sales, and transaction information through a desktop-based management system.
Primary Users	Administrator responsible for managing the system's product, inventory, sales, and transaction information.
Major Deliverable	Working Java Swing desktop application with Login, Dashboard, Products, Inventory, Sales, and Transactions modules, together with project documentation.
Main Constraint	Limited development time, available hardware, student-level resources, and team availability.
Project Success	The core system functions work correctly, data can be stored and retrieved properly, and the administrator can perform the agreed product, inventory, sales, and transaction management processes using the system.

5. REQUIREMENTS GATHERING AND ANALYSIS

5.1 Stakeholders

Stakeholder	Role / Interest
Business Owner / Manager	Approves the project direction and defines what product, inventory, sales, transaction information should be managed.
Administrator	Main current system user who logs in and accesses the management screens.
Sales Staff	Responsible for using the sales-related function of the system to process sales, enter product information, record quantities, and review transaction details.
Cashier	Uses the point of sale workflow to select products, complete checkout, accept payments details, and review permitted transaction information.
Inventory Staff	Inventory staff are responsible for monitoring and managing product and stock information through the Products and Inventory sections of the systems.
Development Team	Analyzes the requirements, designs the interface, develop the system, conducting tests, resolving the errors, and preparing the necessary project documentation.
Customers	Indirect stakeholders who may benefits from a more organized sales process, improved product information management and more efficient transaction processing.

5.2 Requirements Gathering Methods

The development team may use the following methods to collect project requirements:

- Review of the current Alfe Commercial GUI and Java source code.
- Review of the SQL database structure and sample user record.
- Observation of the fields, buttons, labels, and navigation included in each GUI form.

- Follow-up consultation with the project team to confirm future functions.

5.3 Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow an authorized administrator to login using a username and password.
FR-02	The system shall validate that required login fields are not empty.
FR-03	The system shall check entered credentials against the users table.
FR-04	The system shall open the Home page after successful authentication.
FR-05	The system shall provide access to Products, Inventory, Sales, and Transactions from the Home page.
FR-06	The Product interface shall provide fields for product name, brand, category, price, and quantity.
FR-07	The Inventory interface shall support product search, status, and stock update input.
FR-08	The Sales interface shall support product selection, quantity entry, and total related information.
FR-09	The Transaction interface shall display transaction history and transaction details and provide controls for adding, clearing, and searching transaction information.
FR-10	The system shall allow the administrator to logout and return to the Login page.

5.4 Non-Functional Requirements

Category	Sample Requirement
Usability	The system should provide a clear and user friendly interface that allows administrator to perform tasks with minimal difficulty.
Performance	The system should allow users to complete common tasks such as checking products and recording sales without unnecessary delays.
Security	Only the authorized administrator can access the system using login credentials.
Reliability	The system should provide appropriate error messages when invalid information or database problem occur.
Maintainability	The system should allow future improvements or additional features to be added without requiring major changes to the entire application.
Consistency	All screen should follow a similar appearance and arrangement to make the system easier to understand and use.

Data Integrity	Information entered into the system should be correctly stored, retrieved, and update to avoid inaccurate product, sale, and inventory records.

5.5 Sample User Stories and Acceptance Criteria

User Story	Acceptance Criteria
As an administrator, I want to login so that unauthorized users cannot open the management system.	The system should require a username and password and should only continue when the credentials are valid.
As an administrator, I want to open Products so that I can manage the products information.	The Products screen should display relevant product information and available management controls.
As an administrator, I want to check inventory so that I can monitor the stock status.	The system should display the current product stock and quantity so the administrator can easily monitor available inventory.
As an administrator, I want to review transaction so that I can check the transaction details.	The Transaction screen should display available transaction records and their important details.
As an administrator, I want to logout so that the other people cannot access my session.	The system should provide a logout function and return the user to the login screen after confirmation.

6. DEVELOPMENT METHODOLOGY, SDLC MODEL, AND SOFTWARE TOOLS

6.1 Development Methodology Used

The project will use an Agile Development Methodology in developing the ALFE Commercial Sales and Inventory Management System. Agile is appropriate for the project because it allows the researchers to develop the system in manageable parts, test each function, identify problems, and make improvements throughout the development process.

The development team will divide the system into different tasks and modules, such as the Login, Dashboard, Products, Inventory, Sales, and Transactions modules. The team will **regularly** monitor the progress of each task and make necessary adjustments based on testing and feedback.

6.2 SDLC Model Used

The selected SDLC model for the ALFE Commercial Sales and Inventory Management System is the Agile SDLC. Instead of completing the entire system in one long development cycle, the project will be developed gradually through repeated iterations. Each iteration may include planning, requirements analysis, system design, development, testing, and review.

The team will develop the system's main functions in manageable parts, such as the Login, Dashboard, Products, Inventory, Sales, and Transactions modules. After each iteration, the team can identify errors, evaluate the completed functions, and make necessary improvements. Feedback from the project adviser, instructor, or intended user may also be considered before proceeding to the next iteration.

This approach allows the team to monitor progress continuously and make changes when necessary while keeping the development focused on the project's requirements.

6.3 SDLC Model Figure



Figure 1. Agile SDLC Model Used for the Project

6.4 Software and Development Tools Used

The following software and development tools will be used in developing the ALFE Commercial Sales and Inventory Management System:

Tool/Technology	Purpose
Java	Used as the main programming language for developing the system.
Java Swing	Used to create the graphical user interface (GUI) of the desktop application.
NetBeans IDE	Used for writing, designing, debugging, and managing the Java project.
MySQL	Used as the database management system for storing product, inventory, sales, transaction, and user information.
XAMPP	Used to provide the local development environment and run the MySQL database service.
phpMyAdmin	Used to create, manage, and monitor the MySQL database during development.

Java Swing will be used to create the application's graphical user interface, where users can interact with the system. MySQL will be used to store and manage the project's data, while phpMyAdmin will be used mainly for setting up, managing, and maintaining the database during development.

6.5 Reason for Selecting the Methodology and SDLC Model

Agile SDLC with Scrum is appropriate for this project because the system can be developed and improved based on testing, feedback, and the team's progress. Short development cycles allow the team to work on important features first, test the completed functions, identify problems, and make necessary improvements before continuing with the next set of tasks.

This approach also helps the team manage the project's limited academic schedule by dividing the development work into smaller and organized tasks. It allows the team to monitor progress, communicate with one another, and ensure that the main system requirements are completed properly.

6.6 Expected Development Approach

· Identify and prioritize the initial requirements of the ALFE Commercial Sales and Inventory Management System.
· Divide the system into manageable features such as Login, Dashboard, Products, Inventory, Sales, and Transactions.
· Develop and integrate the required features during each sprint.
· Test completed features to identify errors and ensure that they work as expected.
· Present the system's progress to the project adviser or instructor for evaluation and feedback.
· Improve and refine the system based on testing results and accepted feedback.

7. SDLC STAGES AND PROJECT APPLICATION

SDLC Stage	Application to the ALFE Commercial Sales and Inventory Mangement System
Planning	The team will identify the purpose of the ALFE Commercial Sales and Inventory Management System, determine the main objectives, identify the intended users, define the project scope, and consider the available time, hardware, software, and team resources.
Requirements Analysis	The team will identify the information and functions needed for the system, including login, product management, inventory management, sales recording, and transaction management. The team will also determine the functional and non-functional requirements that the system needs to satisfy.
System Design	The team will design the layout and navigation of the Login, Dashboard, Products, Inventory, Sales, and Transactions screens. The team will also plan how the system will store and organize information in the MySQL database.
Development	The team will develop the desktop application using Java and Java Swing in NetBeans. The team will implement the login, product management, inventory, sales, and transaction functions and connect the application to the MySQL database
Testing	The team will test the actual functions of the system, including login, adding and managing products, updating inventory, recording sales, viewing transactions, database operations, and navigation between modules. Errors found during testing will be corrected before the system is finalized.
Deployment	The team will prepare the completed application for use on the intended computer. The

	required Java environment and MySQL database will be configured so that the system can run properly.
Maintenance	The team will correct errors discovered after testing or during system use and make necessary improvements to the existing functions. Database and system information will also be maintained to keep the application working properly

8. PROJECT RISK MANAGEMENT

8.1 Possible Risks Discovered

The team identified possible risks that may affect the development and completion of the ALFE Commercial Sales and Inventory Management System. These risks are related to the project's schedule, system quality, data, team availability, and scope.

8.2 Risk Assessment and Response

Risk	Probability	Impact	Response / Mitigation
Client or user changes important system requirements during development.	Medium	High	Confirm requirements during each sprint and record approved changes before development continues.
Development takes longer than planned.	Medium	High	Prioritize core features, monitor weekly progress, and adjust lower-priority tasks when necessary.
Project files or database information are lost.	Low	High	Maintain regular backups of important project files and database information.
A team member becomes unavailable.	Medium	Medium	Document completed work, share necessary project files among team members, and assign backup responsibilities.
Target computer has limited hardware capability.	Medium	Medium	Test the application on hardware similar to the computer where the system will be used.
Users find some system functions or navigation confusing.	Medium	Medium	Conduct early testing and improve unclear screens or functions based on user and instructor feedback.

9. PROJECT TIME MANAGEMENT AND SCHEDULE

9.1 Major Project Activities

The project schedule outlines the major activities that the team will complete during the development of the ALFE Commercial Sales and Inventory Management System. The activities are arranged according to their expected order, duration, dependencies, and major milestones to help the team monitor project progress and complete the project within the available academic schedule.

9.2 Sample Project Schedule

Activity	Duration	Target Period	Main Output
Planning and Project Study	1 week	Week 1	Project objectives, scope, and initial requirements
Requirements Gathering and Analysis	1 week	Week 2	Confirmed functional and non-functional requirements
System and UI Design	2 weeks	Weeks 3–4	Interface layouts, navigation flow, and database design
Sprint 1 Development	2 weeks	Weeks 5–6	Login, Dashboard, and Products module
Sprint 2 Development	2 weeks	Weeks 7–8	Inventory, Sales, and Transactions modules
Sprint 3 Development	2 weeks	Weeks 9–10	Integration, search/filter functions, validation, and interface improvements
Testing and Revision	1 week	Week 11	Tested system, corrected errors, and improved functions
Deployment and Final Documentation	1 weeks	Weeks 12	Prepared system and completed final documentation

9.3 Project Milestones

Milestone	Target
Requirements approved	End of Week 2

Initial design approved	End of Week 4
Core sales and inventory functions working	End of Week 8
Feature-complete version	End of Week 10
Testing completed	End of Week 11
Final system and presentation prepared	End of Week 12

10. PROJECT BUDGET AND COSTING

10.1 Estimated Project Cost

The project budget includes the resources and expenses that may be needed during the development, testing, documentation, and presentation of the ALFE Commercial Sales and Inventory Management System. The main development software and database tools are available without additional software licensing costs.

Cost Item	Basis	Estimated Cost
Development Software	Java, NetBeans, and Java Swing — free to use	PHP 0
Database Software	MySQL, XAMPP, and phpMyAdmin — free to use	PHP 0
Internet / Data	Internet access for development, research, and communication	PHP 1,000
Transportation	Travel related to meetings, testing, and project activities	PHP 500
Printing / Documentation	Drafts, final documents, and presentation materials	PHP 500
Backup Storage	USB or other storage for project files and database backups	PHP 500
Miscellaneous	Other small project-related expenses	PHP 500
TOTAL ESTIMATED COST		PHP 3,000

10.2 Cost Justification

The estimated budget focuses on the expenses that may be needed during the development and completion of the ALFE Commercial Sales and Inventory Management System. The development and database software have no additional licensing costs because Java, NetBeans, Java Swing, MySQL, XAMPP, and phpMyAdmin are available for use without purchasing software licenses.

The estimated internet and transportation costs will support communication, research, meetings, and project-related activities. Printing and documentation expenses will cover project drafts, final documents, and presentation materials. Backup storage will help protect important project files and database information. The miscellaneous allowance is included for other small and unexpected project expenses.

The budget does not include payment for student labor because the system is being developed as an academic project. The actual expenses may vary depending on the resources already available to the team and the requirements of the project.

11. SCRUM / AGILE APPLICATION

11.1 Scrum Roles

Scrum Role	Project Responsibility
Product Owner	Represents the needs of the intended user, helps prioritize the system requirements, and confirms whether completed features meet the agreed project requirements.
Scrum Master	Organizes Scrum activities, monitors the team's progress, helps address problems or blockers, and keeps the team focused on the sprint goals.
Development Team	Designs, codes, tests, documents, and improves the ALFE Commercial Sales and Inventory Management System, including the Login, Dashboard, Products, Inventory, Sales, and Transactions modules.

11.2 Sample Product Backlog

Priority	Backlog Item	Expected Result
High	User login	Administrator can securely access the system using valid login

		credentials.
High	Product Management	Administrator can add, view, edit, and delete product information.
High	Inventory Management	Administrator can monitor and update product stock information.
High	Sales Recording	Administrator can record sales, selected products, quantities, and total amounts.
High	Transaction Management	Administrator can view and manage recorded transaction information.
Medium	Dashboard	System displays important information such as total products, total sales, and low-stock information.
Medium	Product Search and Filtering	Administrator can search for products and filter them by category.
Medium	Input Validation	System helps prevent incorrect or invalid information from being entered.
Low	Interface Improvements	Additional improvements to the system's layout, navigation, and usability may be completed if time allows.

11.3 Sample Sprint Plan

Sprint	Sprint Goal	Main Work
Sprint 1	Create the basic system structure and access.	Login, Dashboard layout, database connection, and basic navigation.
Sprint 2	Complete product and inventory management.	Product management, product search and filtering, inventory monitoring, and stock updates.
Sprint 3	Complete sales and transaction functions.	Sales recording, quantity and total calculations, saving sales data, and transaction management.
Sprint 4	Prepare the system for final use.	Integration, testing, bug fixes, input validation, interface improvements, and final documentation.

11.4 Scrum Events

Scrum Event	How the Team Will Apply It
Sprint Planning	Select the ALFE system features to be developed during the sprint, prioritize the tasks, and agree on the sprint goal.

Daily Scrum / Team Check-in	Briefly discuss each member's completed work, current tasks, development progress, and any problems or blockers.
Sprint Review	Demonstrate the completed system features, such as Products, Inventory, Sales, or Transactions, to the project group, instructor, or intended user and gather feedback.
Sprint Retrospective	Discuss what worked well, what problems were encountered during development, and what the team can improve in the next sprint

11.5 Simple Scrum Progress Tracking

The team will use a simple task board with three basic columns: **To Do, In Progress, and Done**. Each task or backlog item will be assigned to a team member and updated based on its current status. The team will regularly check the progress of tasks during team check-ins. At the end of each sprint, the team will compare the planned tasks with the completed tasks and use the results to improve the planning and task assignments for the next sprint.

12. PROJECT SUMMARY

The proposed **ALFE Commercial Sales and Inventory Management System** is intended to provide a more organized way of managing product information, inventory, sales, and transaction records through a desktop-based application. The project focuses on the core functions needed for product, inventory, sales, and transaction management, which helps keep the project scope realistic and manageable for a student development team.

Based on the project feasibility review, the system can be developed using commonly available software and hardware, including Java, Java Swing, NetBeans, MySQL, XAMPP, and phpMyAdmin. Using Agile and Scrum allows the team to develop the system in manageable parts, test features during development, and make improvements based on testing and feedback. The identified risks, project schedule, and estimated budget also provide the team with a structured way to monitor and manage the project throughout its development.

OPTIONAL ATTACHMENTS

The instructor may allow the following supporting materials when applicable:

· Client interview guide or summarized interview notes

· Process flow or simple workflow diagram

· Wireframes / user interface mockups

· Gantt chart or detailed project schedule

· Screenshots of Scrum board or task tracker

· Prototype screenshots

Sample note: *These attachments support the main documentation. They should not replace the explanations required in Sections 1 to 12.*